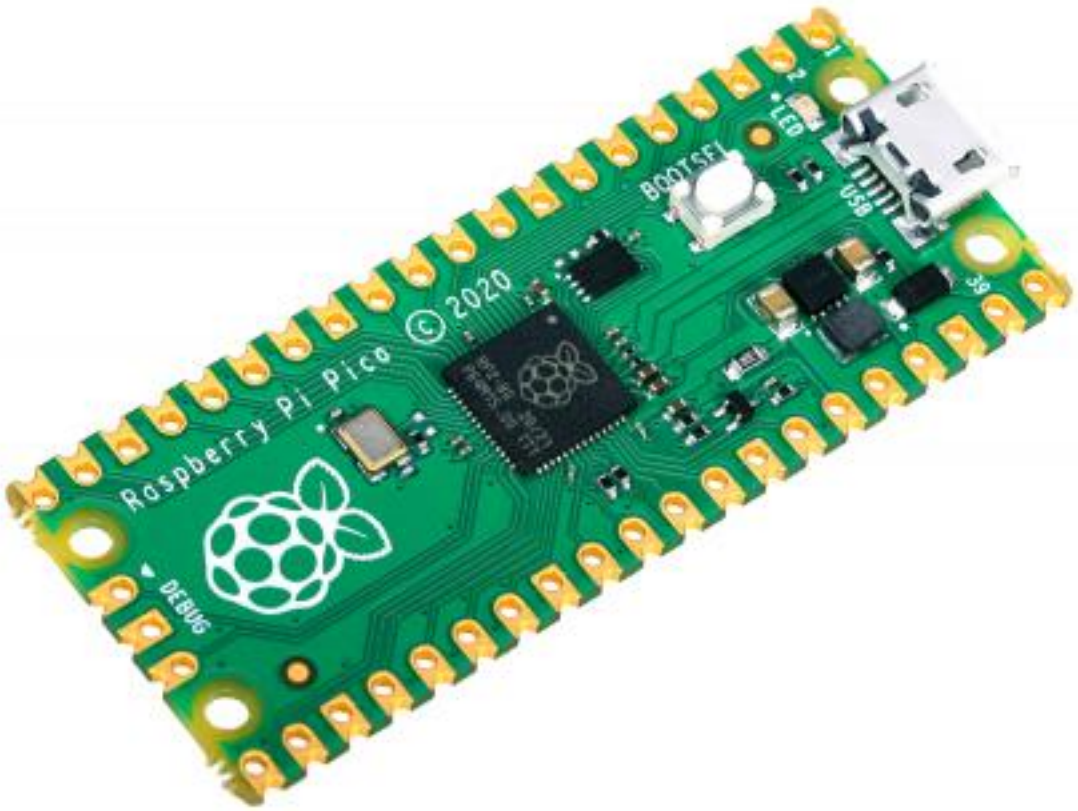
Raspberry pi Pico



RASPBERRY PI



Raspberry Pi

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What is Raspberry-pi

- **What is a Raspberry Pi?** Raspberry pi is the name of the “credit card-sized computer board” developed by the Raspberry pi foundation, based in the U.K.
- Can be plugged into TV / Monitor for a fully functional computer capability.
- Raspberry Pi is a series of powerful, small single-board computers.
- Raspberry Pi is launched in 2012 and there have been several iterations and variations released since then.
- All versions consist of a Broadcom system on a chip (SoC) with an integrated ARM-compatible CPU and on-chip graphics processing unit (GPU).
- The original device had a single-core Processor speed of device ranges from 700 MHz to 1.2 GHz and a memory range from 256 MB to 1 GB RAM.
- To store the operating system and program memory Secure Digital (SD) cards are used. Raspbian OS which is a Linux operating system is recommended OS by Raspberry Pi Foundation.

Raspberry Pi Diagram

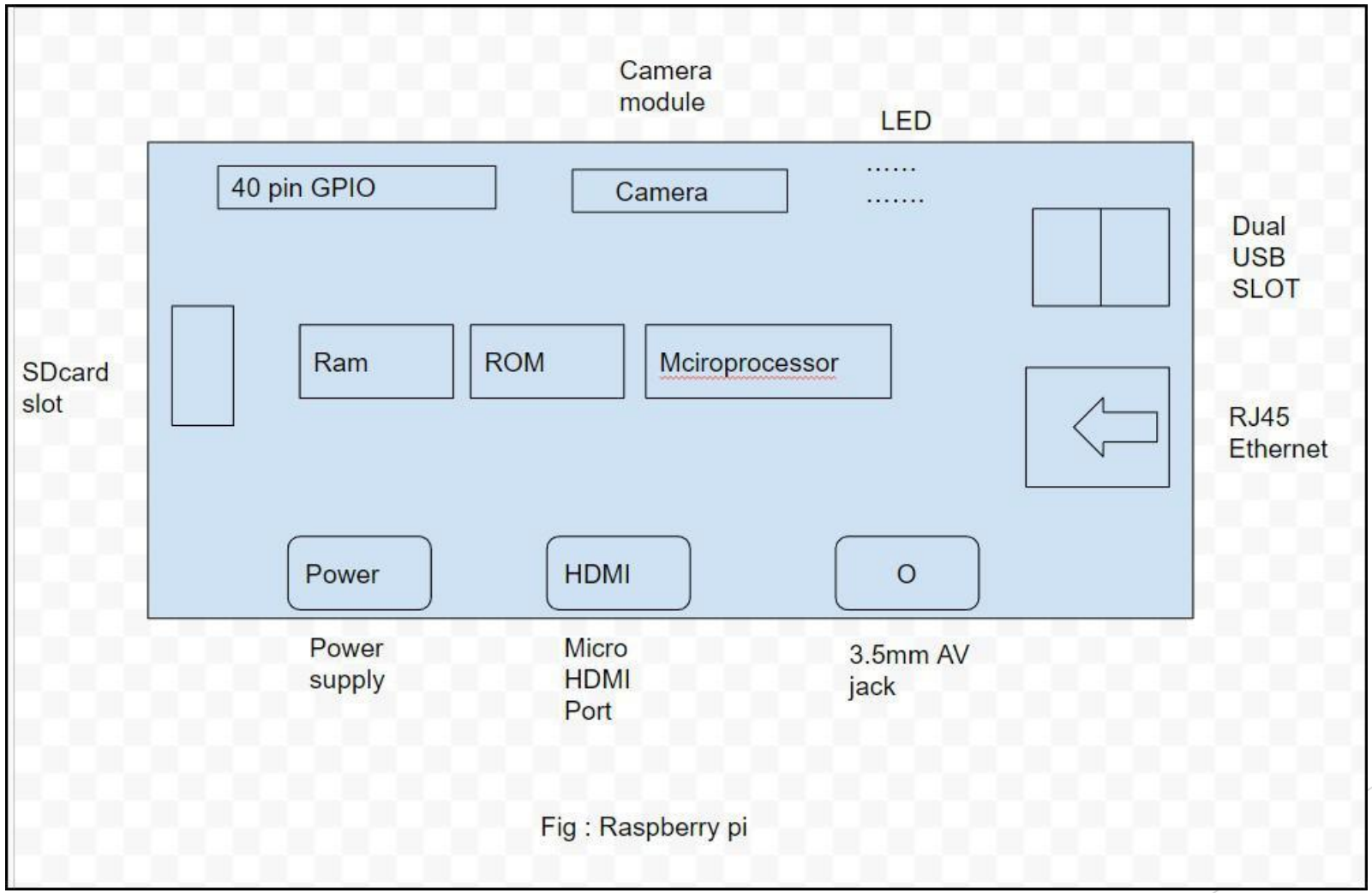


Fig : Raspberry pi

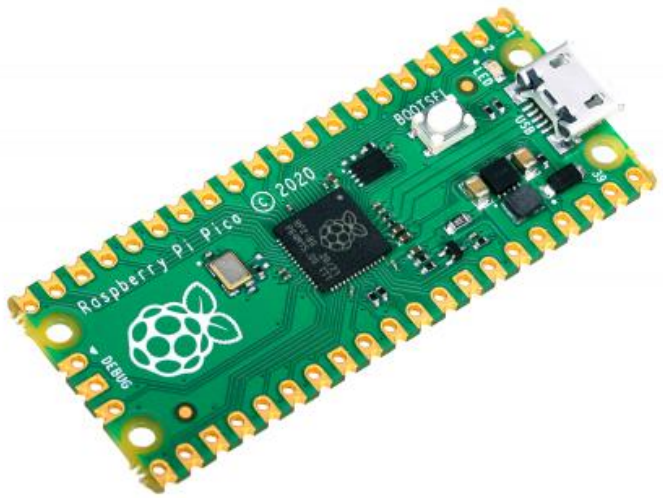
Version's of Raspberry Pi

Raspberry Pi Model	Released	USB Ports	CPU	RAM	Bluetooth	WiFi
Raspberry Pi 1 B	Feb 2012	2x USB 2.0	700MHz	512MB	No	No
Raspberry Pi 1 B+	July 2014	4x USB 2.0	700MHz	512MB	No	No
Raspberry Pi 1 A+	Nov 2014	1x USB	700MHz	512MB	No	No
Raspberry Pi 2	Feb 2015	4x USB	900MHz	1GB	No	No
Raspberry Pi Zero	Nov 2015	1x Micro USB	1GHz	512MB	No	No
Raspberry Pi 3	Feb 2016	4x USB	1.2GHz	1GB	4.1 LE	Yes
Raspberry Pi Zero W	Feb 2017	1x Micro USB	1GHz	512MB	4.1	Yes
Raspberry Pi 3 B+	Mar 2018	4x USB 2.0	1.4GHz	1GB	4.2, BLE	Yes
Raspberry Pi 3 A+	Nov 2018	1x USB 2.0	1.4GHz	512MB	4.2, BLE	Yes
Raspberry Pi 4	June 2019	2X USB 3.0, 2X USB 2.0	1.5GHz	1GB, 2GB, 4GB, or 8GB	5.0, BLE	Yes
Raspberry Pi 400	November 2020	2X USB 3.0, 2X USB 2.0	1.8GHz	4GB	5.0, BLE	Yes
Raspberry Pi Pico	January 2021	USB C	133MHz	264KB	No	No

Raspberry Pi Variants









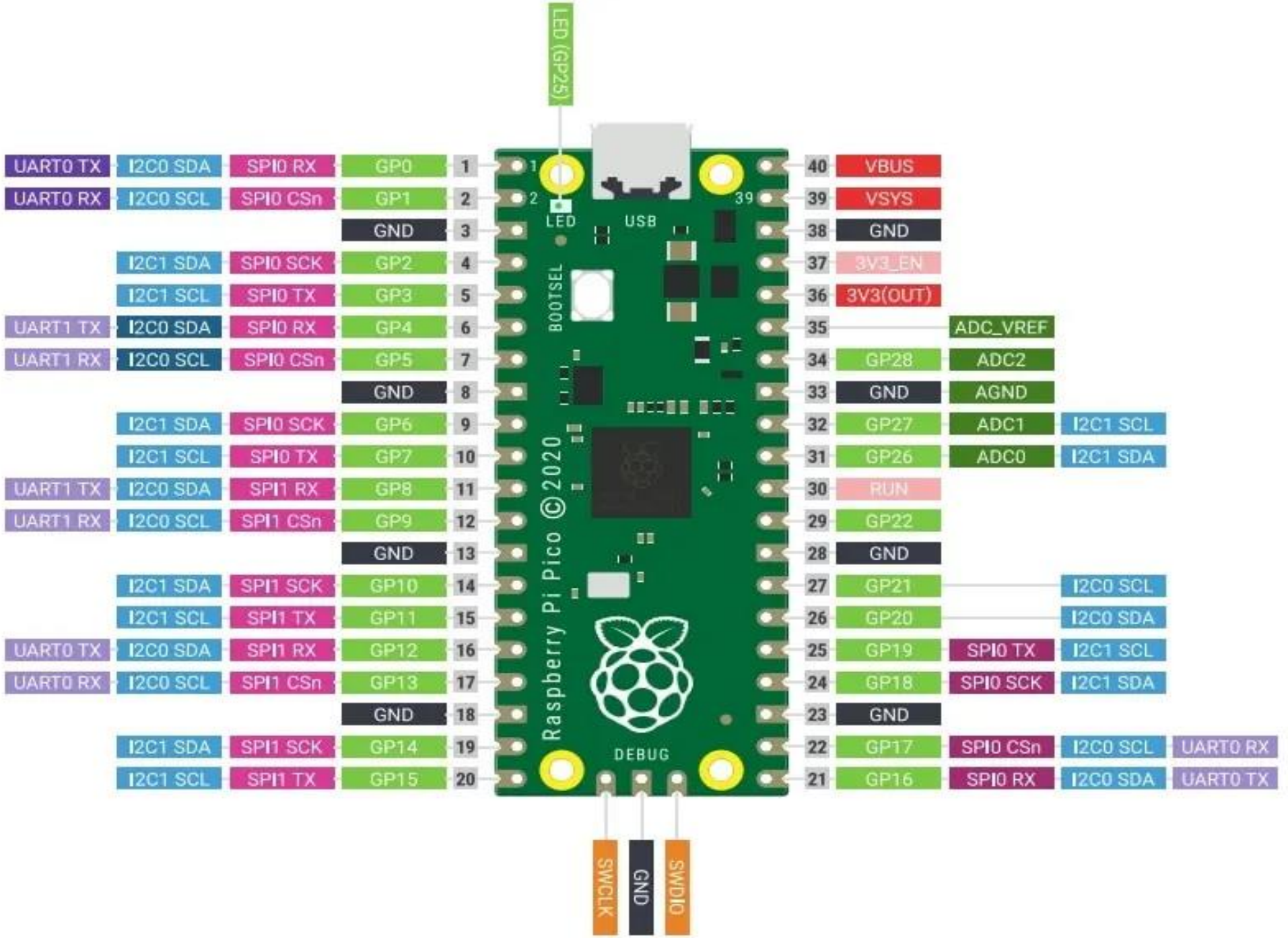
Differences between pi & pi pico

Raspberry PI	Raspberry PI PICO
❖ A Raspberry Pi is a Single Board Computer that **runs Linux.** ❖ You can connect a monitor, keyboard and mouse and interact with via a terminal prompt or a graphical user **interface** such as the Raspberry Pi OS desktop. ❖ You can run many different **Linux applications** on a Raspberry Pi and it has a ton of processing power and RAM.	❖ Raspberry Pi PICO's are champion multi-taskers, microcontrollers shine in projects where they have one job and need to do it well. ❖ Because they're programmed from another computer and don't need to run a full OS, they have greatly reduced power consumption compared to SBCs and are ideal for powering from batteries. ❖ Programs on microcontrollers just start running once they're powered up, so there's no boot up time, and you usually don't need to shut them down safely. Used in industrial machines, coffee makers etc..,

Difference between Arduino & Pi

S No.	Arduino	Raspberry Pi
1.	In the year **2005**, the classrooms of the Interactive Design Institute in Ivrea, Italy, first introduced the Arduino board.	In the year **2012**, Eben Upton first introduced the Raspberry Pi device in February.
2.	Control unit of the Arduino is from the **Atmega family.**	The control unit of Raspberry Pi is from the **ARM family.**
3.	Arduino is based on a **microcontroller**.	While Raspberry Pi is based on a **microprocessor**.
4.	It is designed to **control the electrical components** connected to the circuit board in a system.	While Raspberry Pi **computes data and produces valuable outputs,** and controls components in a system based on the outcome of its computation.
5.	Arduino boards have a **simple hardware and software structure.**	While Raspberry Pi boards have **a complex architecture of hardware and software**.

Microcontroller	Microprocessor
A microcontroller is a specialized form of a microprocessor	The microprocessor is designed to be general-purpose.
It is cost-effective.	It is a silicon chip
It is self-sufficient.	It is a dependent unit
The microcontroller is used to perform a particular tasks.	The Microprocessor is used to perform a certain task.
Its power consumption is low.	Its power consumption is high.
It contains CPU, RAM, ROM, Registers, Timer and input/output ports.	It requires a combination of timers, controllers memory chips.


LED (GP25)
LED
USB
BOOTSEL
DEBUG
Raspberry Pi Pico ©2020
SWCLK
GND
SWDIO
UART0 TX I2C0 SDA SPI0 RX GP0 1
UART0 RX I2C0 SCL SPI0 CSn GP1 2
GND 3
I2C1 SDA SPI0 SCK GP2 4
I2C1 SCL SPI0 TX GP3 5
UART1 TX I2C0 SDA SPI0 RX GP4 6
UART1 RX I2C0 SCL SPI0 CSn GP5 7
GND 8
I2C1 SDA SPI0 SCK GP6 9
I2C1 SCL SPI0 TX GP7 10
UART1 TX I2C0 SDA SPI1 RX GP8 11
UART1 RX I2C0 SCL SPI1 CSn GP9 12
GND 13
I2C1 SDA SPI1 SCK GP10 14
I2C1 SCL SPI1 TX GP11 15
UART0 TX I2C0 SDA SPI1 RX GP12 16
UART0 RX I2C0 SCL SPI1 CSn GP13 17
GND 18
I2C1 SDA SPI1 SCK GP14 19
I2C1 SCL SPI1 TX GP15 20
40 VBUS
39 VSYS
38 GND
37 3V3_EN
36 3V3(OUT)
35 ADC_VREF
34 GP28 ADC2
33 GND AGND
32 GP27 ADC1 I2C1 SCL
31 GP26 ADC0 I2C1 SDA
30 RUN
29 GP22
28 GND
27 GP21 I2C0 SCL
26 GP20 I2C0 SDA
25 GP19 SPI0 TX I2C1 SCL
24 GP18 SPI0 SCK I2C1 SDA
23 GND
22 GP17 SPI0 CSn I2C0 SCL UART0 RX
21 GP16 SPI0 RX I2C0 SDA UART0 TX

Specifications of pi pico

Type	Component
Processor	RP2040, 2x Arm Cortex-M0+ (133 MHz)
Memory	264 KB SRAM
Storage	2 MB flash memory
Connectivity	2x 20 GPIO pins (20 on each side), as: - 26x multifunction GPIO Pins - 2x UART - 2x SPI - 2x I2C controllers - 8x programmable I/IO (PIO) - 8 GND pins - 3 power pins VCC
Power	1x Micro USB Support 1.8 to 5.5V DC
Dimensions	21 mm x 51 mm

Pico PIN Explanations

- UART = Universal Asynchronous Receiver-Transmitter
- I2C stands for **Inter-Integrated Circuit.**
- **Serial Data (SDA)** - Transfer of data takes place through this pin.
- **Serial Clock (SCL)** - It carries the clock signal. SCL **controls the timing of the data transmission**. It is a master-slave communication protocol, where the Raspberry Pi Pico acts as the master
- SPI = Serial Peripheral Interface
- The primary SPI pins on the Raspberry Pi Pico are as follows: GP16 (RX):Transmitter GP17 (CS):Chip Select These pins are used to **connect the Raspberry Pi Pico W to the peripheral device for SPI communication**.
- Serial Clock In GP18 (SCK)
- GP19 (TR): Transmittter.
- As data is received (RX) on the input and data is sent (transmitted, TX) on the output, from the master's point of view, MISO. MOSI is reversal of MISO.

Blinking inbuilt LED

```
import machine
import utime
led_onboard = machine.Pin(25, machine.Pin.OUT)

while True:
    led_onboard.value(1)
    utime.sleep(1)
    led_onboard.value(0)
    utime.sleep(0.5)
```

Running LED

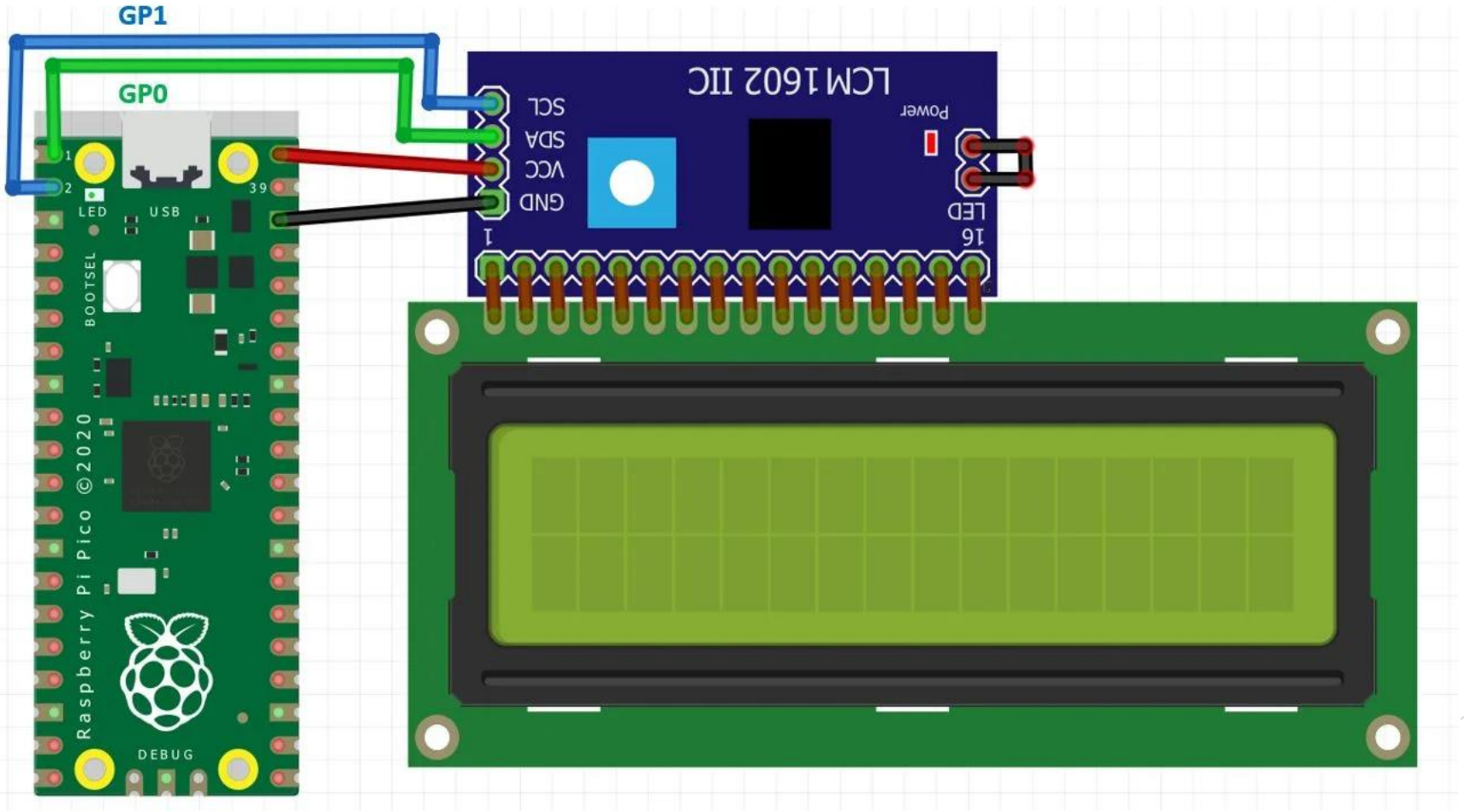
```
from machine import Pin
from time import sleep

leds = [1,2,3,4,5]

for led in leds:
    led_pin = Pin(led,Pin.OUT)

while True:
    for i in range(0,6,1):  # starting at 0 up until 6 with increment by 1(0,6,1)
        led_pin.value(1)
        sleep(0.05)
        led_pin.value(0)
```

Pi pico with LCD – i2c



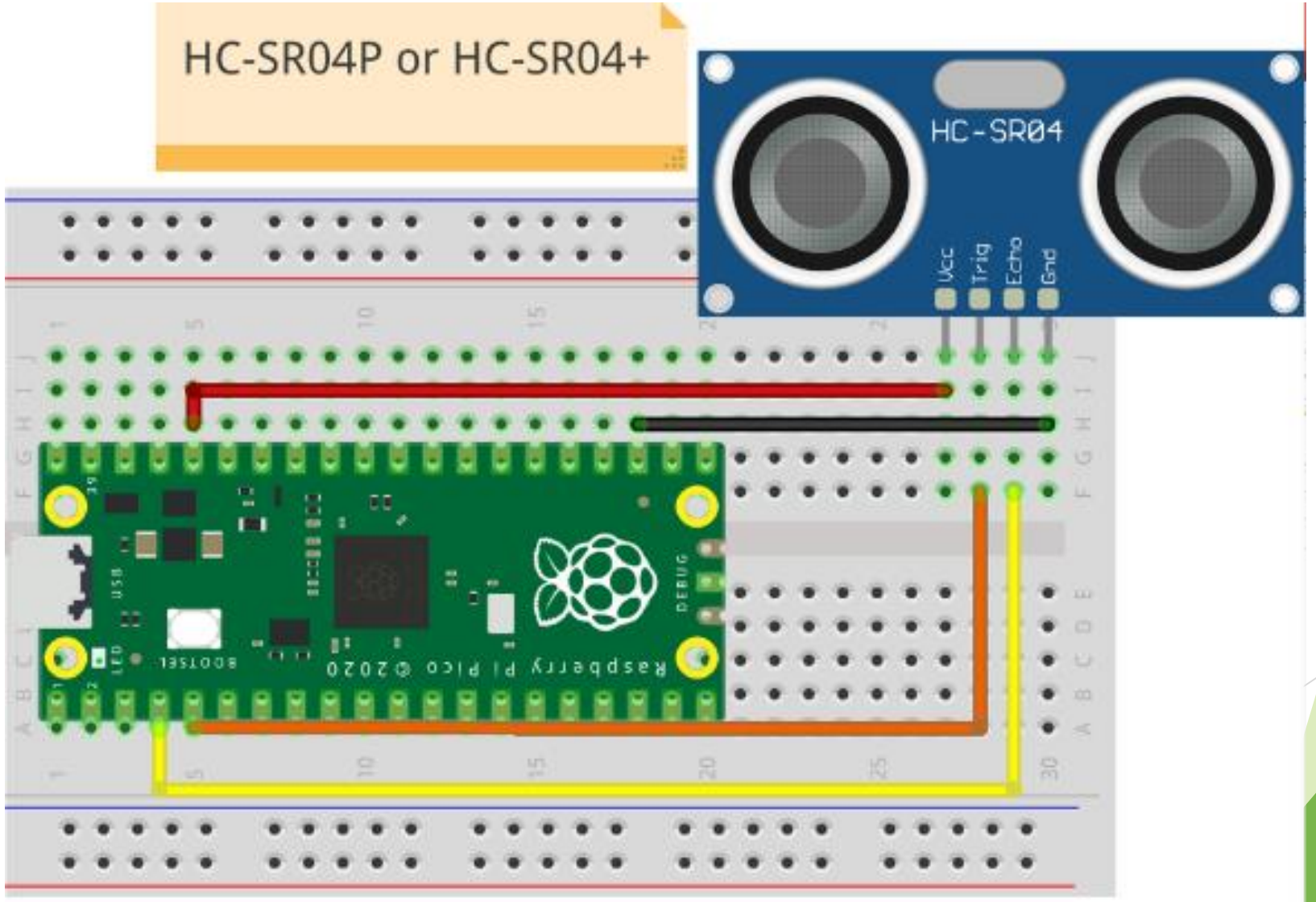
```
import machine
from machine import I2C
from lcd_api import LcdApi
from i2c_lcd import I2cLcd
from time import sleep

I2C_ADDR = 0x27
totalRows = 2
totalColumns = 16

i2c = I2C(0, sda=machine.Pin(0), scl=machine.Pin(1), freq=400000)
lcd = I2cLcd(i2c, I2C_ADDR, totalRows, totalColumns)

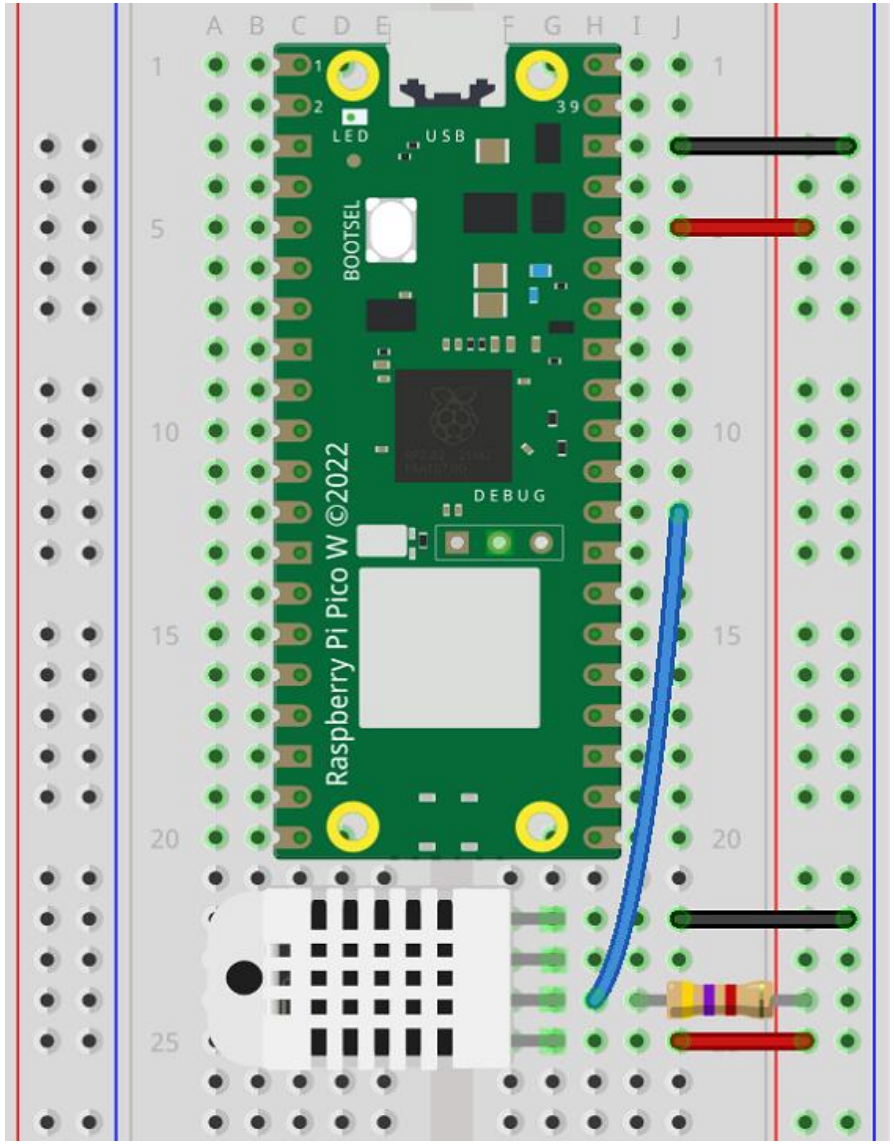
while True:
    lcd.putstr("Lets Count 0-10!")
    sleep(2)
    lcd.clear()
    for i in range(10):
        lcd.putstr(str(i))
        sleep(1)
 lcd.clear()
```

Pi pico with ultrasonic



```
from machine import Pin
import utime
trigger = Pin(3, Pin.OUT)
echo = Pin(2, Pin.IN)
def ultra():
    trigger.low()
    utime.sleep_us(2)
    trigger.high()
    utime.sleep_us(5)
    trigger.low()
    while echo.value() == 0:
        signaloff = utime.ticks_us()
    while echo.value() == 1:
        signalon = utime.ticks_us()
    timepassed = signalon - signaloff
    distance = (timepassed * 0.0343) / 2
    print("The distance from object is ",distance,"cm")
while True:
    ultra()
    utime.sleep(1)
```

Pi pico with Temperature/ Humidity Sensor



```
from machine import Pin
from time import sleep
import dht

sensor = dht.DHT22(Pin(22))

#sensor = dht.DHT11(Pin(22))

while True:
  try:
    sleep(2)
    sensor.measure()
    temp = sensor.temperature()
    hum = sensor.humidity()
    temp_f = temp * (9/5) + 32.0
    print('Temperature: %3.1f C' %temp)
    print('Temperature: %3.1f F' %temp_f)
    print('Humidity: %3.1f %%' %hum)
  except OSError as e:
    print('Failed to read sensor.')
```